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NMAM INSTITUTE OF TECHNOLOGY, NITTE

(An Autonomous Institution affiliated to VTU, Belagavi)

Third Semester B.E. (ISE) (Credit System) Degree Examinations

Supplementary Examinations – December 2025

21IS304 – OBJECT ORIENTED PROGRAMMING

Duration: 3 Hours

Max. Marks: 100

*Note: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II** each and **One full** question from **Unit – III**.*

Unit – I		Marks	BT*	CO*	PO*
1. a)	What is meant by byte code? Briefly explain how Java is platform independent.	6	L*2	1	1
b)	Describe the structure of a typical Java program with an example.	7	L2	1	2
c)	Explain the process of defining and creating a package with suitable examples.	7	L2	2	1
2. a)	What are difference between Methods and Constructor?	4	L1	1	2
b)	Explain the significance of public, protected and private access specifiers in inheritance.	6	L2	1	1
c)	Write a program to demonstrate hierarchical and multiple inheritance using interfaces.	10	L3	1	2
3. a)	How to design and implement an interface in Java? Give an example.	6	L2	2	2
b)	Discuss method overloading and method overriding with example	8	L2	2	1
c)	Discuss the different purpose of using super keyword in java with examples	6	L2	2	2
Unit – II					
4. a)	What is exception handling? Explain of exception handling in the case of division by zero.	8	L2	3	2
b)	Write a java program that illustrates the application of multiple catch statements.	8	L3	3	2
c)	Discuss the use of finally with example.	4	L2	3	2
5. a)	Write a program that creates two threads. First thread prints the numbers from 1 to 100 and the other thread prints the numbers from 100 to 1.	10	L3	3	2
b)	How many ways are possible in java to create multiple threaded programs? Discuss the differences between them.	6	L2	3	1
c)	Can we have an empty catch block? Justify.	4	L2	3	2

P.T.O.

21IS304

Supplementary – December 2025

6. a) What are the different ways to handle exceptions? Explain.	8	L2	3	1
b) Write a program to demonstrate isAlive() and join() methods.	8	L3	3	2
c) Write a short note on Thread priorities.	4	L2	3	1

Unit – III

7. a) What is collection interface? Explain the methods defined by collection interface.	8	L2	4	1
b) What are generics? Demonstrate a simple generics class with an example.	6	L3	4	2
c) What are similarities and difference between ArrayList and Vector? Explain.	6	L2	4	1
8. a) Discuss JavaFX Application skeleton.	8	L2	4	2
b) Write a short note on Stage and Scene class.	4	L2	4	1
c) Demonstrate a JavaFX Label.	8	L3	4	2

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome.
